

Topology General Exam – August 2023

Instructions: This is a four hour exam. Your solutions should be legible and clearly organized, written in complete sentences in good mathematical style on your own paper. All work should be your own-no outside sources are permitted-using methods and results from the first year topology courses. Each problem is worth the same number of points.

Problem 1

Let $x_i, y_i, z, i = 1, \dots, n$, denote the coordinates on $\mathbb{R}^{2n+1}$. Consider the 1-form

$$w = dz + \sum_{i=1}^n x_i dy_i.$$

Compute $w \wedge (dw)^n$ and show that it is a volume form. Here $(dw)^n$ denotes the n -fold wedge product $dw \wedge \dots \wedge dw$.

Problem 2

Consider the subset S of $\mathbb{R}^4$ defined by the equation $x_1^2 + x_2^2 - x_3^2 - x_4^2 = 1$, and consider the function $f : \mathbb{R}^4 \rightarrow \mathbb{R}$ given by $f(x_1, x_2, x_3, x_4) = x_1$.

Show that S is a submanifold of $\mathbb{R}^4$, and identify the critical points and critical values of the function f restricted to S , $f|_S : S \rightarrow \mathbb{R}$.

Problem 3

Let M be a compact manifold with boundary. Prove that there does not exist a smooth map $f : M \rightarrow \partial M$ such that $f(x) = x$ for all $x \in \partial M$.

Problem 4

Let M be a smooth manifold and $f : M \rightarrow M$ a smooth map. Suppose that $p \in M$ is a fixed point of f , meaning $f(p) = p$. Prove that the following conditions are equivalent:

- The graph $\{(x, f(x)) \mid x \in M\}$ of f intersects the diagonal $\{(x, x) \mid x \in M\}$ transversely at the point $(p, p) \in M \times M$.
- 1 is not an eigenvalue of the differential $d_p f : T_p M \rightarrow T_p M$.

Hint. It may be helpful to consider the graph $\{(v, d_p f(v)) \mid v \in T_p M\}$ of the differential $d_p f$ and the conditions under which it is transverse to the diagonal $\{(v, v) \mid v \in T_p M\}$ in $T_p M \times T_p M$.

Problem 5

Suppose that the 3-torus $T^3 = S^1 \times S^1 \times S^1$ is written as a union $T^3 = U_1 \cup \dots \cup U_m$ of open subsets U_j each homeomorphic to $\mathbb{R}^3$. Prove that for some $k \leq m$, the intersection $(U_1 \cup \dots \cup U_{k-1}) \cap U_k$ is either empty or disconnected.

Problem 6

Let $f : S^n \rightarrow S^n$ be a continuous map.

- Prove that if $\deg(f) \neq (-1)^{n+1}$ then f has a fixed point.
- Suppose that $f(x) = f(-x)$ for all $x \in S^n$. Prove that $\deg(f)$ is even, and if in addition n is even, then $\deg(f) = 0$.

Problem 7

Let $X = \mathbb{R}P^2 \vee \mathbb{R}P^2$.

- Prove that X does not admit a connected, regular, 3-sheeted covering space.
- Construct a connected (non-regular) 3-sheeted covering space of X .

Problem 8

Let $K \subset S^3$ be a smooth submanifold diffeomorphic to a circle, $K \cong S^1$ (so K is a "knot" in the 3-dimensional sphere). Use the axioms and properties of homology, including for example the long exact sequence of a pair and the Mayer-Vietoris sequence, to answer the following:

- Find the relative homology groups $H_j(S^3, K)$ for all j .
- Prove that $H_1(S^3 - K) \cong \mathbb{Z}$.